

NSX Firewall Port Matrix

OpenShift Container Platform 4.20 on VMware vSphere (OVN-Kubernetes + HAProxy)

Source	Destination	Port	Purpose
Bastion / Installer	vCenter & ESXi hosts	TCP 443	vCenter API and ESXi access; required for installer and control plane for CSI and machine management.
External Clients & All Nodes	HAProxy VIPs	TCP 6443	Kubernetes API access.
All Nodes	HAProxy VIPs	TCP 22623	Machine Config Server (MCS) access.
External Clients & All Nodes	HAProxy VIPs	TCP 80, 443	Ingress HTTP/HTTPS access.
HAProxy	Control Plane Nodes	TCP 6443, 22623	Load balancer routing to API and MCS backends.
HAProxy	Worker / Router Nodes	TCP 80, 443	Load balancer routing to Ingress backends.
All Nodes	All Nodes	UDP 6081	OVN-Kubernetes Geneve overlay network.
All Nodes	All Nodes	TCP/UDP 9000-9999	Host-level services.
All Nodes	All Nodes	TCP 10250, 10256	Kubelet (10250) and network health (10256).
Control Plane Nodes	Control Plane Nodes	TCP 2379-2380	etcd peer and client communication.
All Nodes	All Nodes	TCP/UDP 30000-32767	NodePort service range.
All Nodes	All Nodes	UDP 500, 4500	IPsec (if OVN IPsec is enabled).
All Nodes	All Nodes	TCP 9100-9101, 9537	Metrics ports for monitoring scraping.
All Nodes	All Nodes	TCP/UDP 53, 5353	Internal DNS and DNS-based components.
All Nodes	Mirror Registry VM	TCP 443 or 8443	Image pulls from the internal mirror registry.
All Nodes	DNS Server	TCP/UDP 53	Forward and reverse DNS resolution.
All Nodes	NTP Server	UDP 123	Time synchronization.

Notes: Based on official OpenShift 4.20 network connectivity requirements. Allow bidirectional traffic for node-to-node ranges unless a more granular design is used. Also consider allowing ICMP for reachability tests, TCP 1936 (router metrics), the full Kubernetes reserved range TCP 10250–10259, and ESP if OVN IPsec is enabled.

Document generated for NSX Distributed Firewall / Gateway Firewall rule planning.